

Overview of the Course

Your teachers:

Schedule :

Two weekly lectures

Monday	09:00	11:00	C1
Thursday	14:00	16:00	A1



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One weekly lecture for experiencing with practical applications

Wednesday	11:00	13:00	C1
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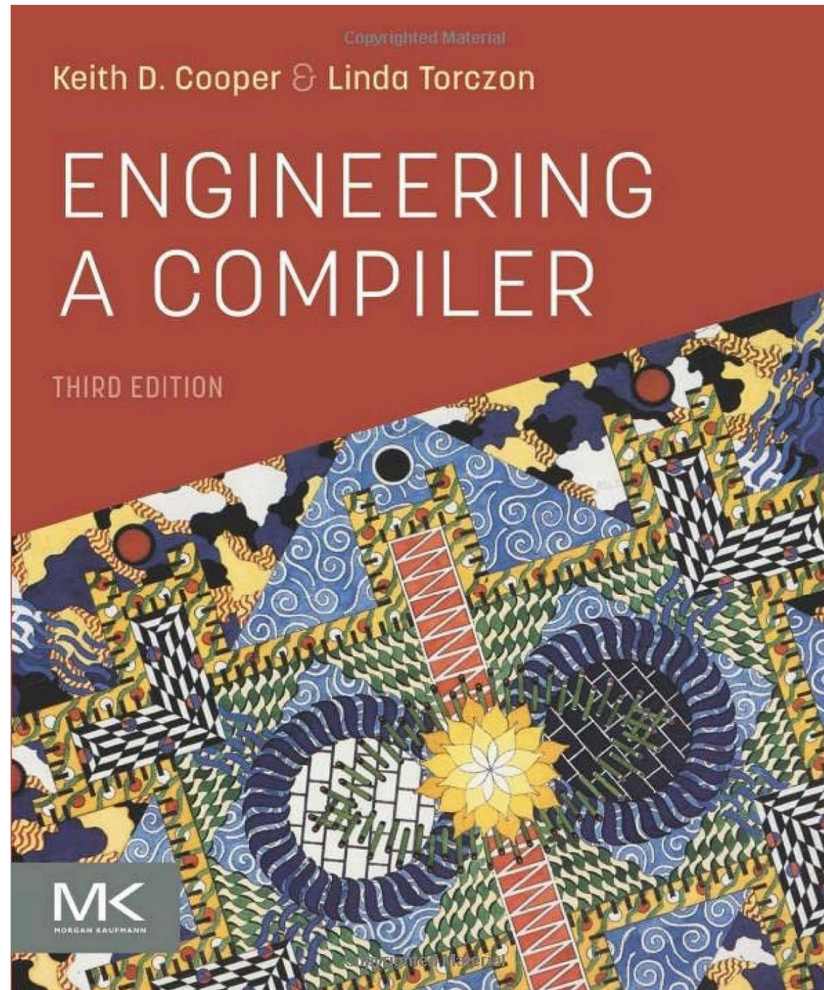


Giuseppe Lomurno
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What we will see

- Formal languages (maybe a recall for someone):
 - Grammars, automata, theorems, regular and context free languages
 - Chomsky hierarchy
- Lexical analysis
- Parser
- Contextual analysis
- Intermediate representation
- Code shape
- Optimization
- Dataflow analysis
- Ownership, Borrowing, Lifetimes, and Non-Lexical Lifetimes (NLL) in Rust
- Register allocation

Our textbook



Other informations

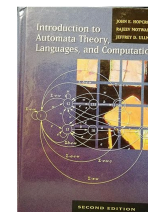
- A web page, I will add there all the slides

www.di.unipi.it/~gori/Linguaggi-Compilatori2025

- A group on Microsoft Team where we can communicate
[Compilation Techniques](#)

Material for specific topics:

- Introduction to Automata Theory, Languages, And Computation.
Hopcroft, Motwani, Ullman



About this teacher

Roberta Gori

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My own research program

Program analysis for verification and optimization

- Static analysis to prove properties on program behaviors
- Abstract interpretation based techniques for proving correctness or devise bugs in programs:
connections with  Meta and real world tools

CAVEAT on the slides

- Apart from the first two weeks, we will follow the slides of the authors of the book
- These are slides for a course in U.S.A. which intend to give you basic concepts and ideas (they do not summarise the book but they give you more ideas and perspectives)
- All the details have to be found in the book
- Use the slides as a guide on the different topics

Final Exam

The exam will consist in

1. a **project** developed partly during the hours of lesson
2. a **seminar** on a chosen topic between a list
3. an **oral** discussion starting from the seminar but covering the topics of the course

Please, actively participate to lectures

Compilers

- What is a **compiler**?
 - A program that takes other programs and prepare them for execution
- In particular, a program that takes a program and translate it in program written in a target language
 - The target language is in general the instruction set of an architecture
 - The target language can be a human-oriented programming language (**Source-to-source translators**)

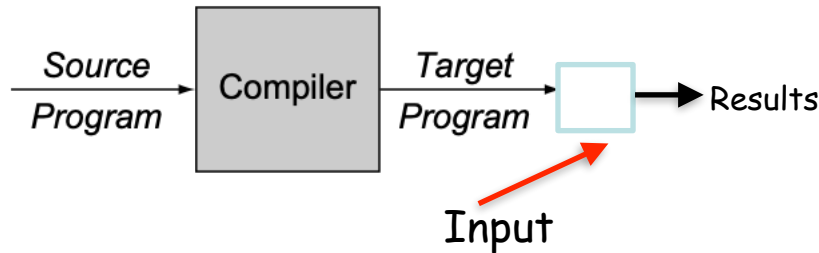
Compilers vs Interpreters

Both compilers and interpreters are programs that translate code written in a high-level programming language into machine code, which the computer can understand.

The main difference is **when** and **how** they translate the code.

Compilers

- A compiler is a program that translates the entire program into machine code before it is executed.



How it works:

1. You write the source code.
2. The compiler translates all the code at once.
3. It produces an executable file.
4. You run the executable file.

Key characteristics:

- Translation happens **once**, before execution.
- The result is a separate **executable program**.
- Execution is usually **faster**.
- Errors are shown **after compilation**, all at once.

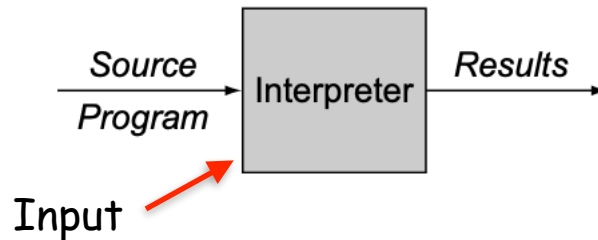
Analogy:

Compilation is like translating a whole book before printing it.

Once translated it, you can read it anytime without translating it again.

Interpreters

- What is an **interpreter**?
 - A program that reads a program and an input and translates the program line by line, at runtime.



How it works:

1. You write the source code.
2. The interpreter reads one line.
3. It translates and executes it immediately.
4. Then it moves to the next line.

Key characteristics:

- Translation happens **during execution**.
- No separate executable file is produced.
- Execution is usually **slower**.
- Errors appear **immediately**, stopping execution.

Analogy:

An interpreter is like a live translator who translates each sentence as someone speaks.

Compilers vs Interpreters

Translates whole program at once	Translates line by line
Creates executable file	No separate executable
Faster execution	Slower execution
Errors shown after compilation	Errors shown immediately

It's a property of the implementation !

- C and C++ are typically compiled,
- Scheme is typically interpreted
- Python is typically interpreted but:
 - CPython compiles source code into bytecode and then interprets that bytecode
 - PyPy uses Just-In-Time (JIT) compilation
 - Cython or Nuitka translate Python into compiled C code
- Java: both compiled and interpreted
 - javac compiles source code into bytecode
 - The JVM may interpret the bytecode
 - Or it may JIT-compile it into native machine code

AOT vs JIT

Both **AOT (Ahead-Of-Time)** and **JIT (Just-In-Time)** are techniques used to translate program code into machine code (compilation).

The main difference is **when** the translation happens.

AOT - Ahead-Of-Time Compilation

With AOT, the program is compiled before it is executed.

How it works:

1. You write the source code.
2. The compiler translates it into machine code in advance.
3. An executable file is created.
4. You run the executable.

Key characteristics:

- Compilation happens before runtime.
- Produces a standalone executable.
- Fast startup time.
- Performance is predictable.
- No compilation overhead while running.

Analogy:

AOT is like cooking a meal completely before serving it. When it's time to eat, everything is already ready

JIT - Just-In-Time Compilation

With JIT, the program is compiled during execution.

Instead of compiling everything in advance, the system compiles parts of the program while it runs, usually optimizing the most frequently executed parts.

How it works:

1. The program starts running.
2. Code is initially interpreted or partially compiled.
3. Frequently used parts are compiled into optimized machine code during runtime.
4. Performance improves as the program runs.

Key characteristics:

- Compilation happens at runtime.
- Slower startup time.
- Can achieve very high performance after optimization.
- Adapts to actual program behavior.

Analogy:

JIT is like starting to cook while guests are already eating — and improving the recipe each time you prepare it again.

Why Study Compilers?

Deep Understanding of Programming Languages

Studying compilers provides a deeper insight into how programming languages work, including structures, optimizations, and resource management.

Performance Optimization

You will learn how to optimize code for speed and memory usage, crucial for high-performance applications or resource-constrained environments.

Foundation for Language Development

Knowledge of compilers is essential for creating new programming languages or extending existing ones.

Development Tools and Automation

Compilers are the backbone of many tools like IDEs, debuggers, and static analyzers, enabling productivity improvements for developers.

Problem-Solving and Transferable Skills

Compilers involve complex algorithmic and data structure, applicable in various fields like OS development or search engines.

Research Opportunities

Compiler technologies, especially in modern areas like parallel programming and code analysis, offer rich research possibilities.

Still many open problems!

Compiler are interesting

- Compiler construction involves ideas from many different parts of computer science

Artificial intelligence	Greedy algorithms Heuristic search techniques
Algorithms	Graph algorithms, union-find Dynamic programming
Theory	DFAs & PDAs, pattern matching Fixed-point algorithms
Systems	Allocation & naming, Synchronization, locality
Architecture	Pipeline & hierarchy management Instruction set use

Performance: reducing the price of language abstraction

Computer Science is the art of creating virtual objects and making them useful.

- We invent abstractions and uses for them
- Programming is the way we realize these inventions

Well written compilers make such abstraction affordable

- Cost of executing code should reflect the underlying work rather than the way the programmer chose to write it
- Change in expression should bring small performance change
- Cannot expect compiler to devise better algorithms
 - Don't expect bubblesort to become quicksort

Making Languages Usable

It was our belief that if FORTRAN, during its first months, were to translate any reasonable "scientific" source program into an object program only half as fast as its hand-coded counterpart, then acceptance of our system would be in serious danger... I believe that if we failed to produce efficient programs, the widespread use of languages like FORTRAN would have been seriously delayed.

— John Backus on the subject of the 1st FORTRAN compiler

Era nostra convinzione che se FORTRAN, nei suoi primi mesi di vita, avesse tradotto qualsiasi ragionevole programma sorgente "scientifico" in un programma oggetto che una volta eseguito fosse piu' veloce solo la metà della codifica a mano dello stesso programma, l'accettazione del nostro sistema di compilazione sarebbe stata in serio pericolo... Credo che se non fossimo riusciti a produrre programmi compilati efficienti, l'uso diffuso di linguaggi come il FORTRAN sarebbe stato seriamente ritardato.

Simple Examples

Which is faster?

```
for (i=0; i<n; i++)  
  for (j=0; j<n; j++)  
    A[i][j] = 0;
```

All three loops have distinct
performance.

0.51 sec on 10,000 x 10,000 array
 $\approx 5 \times 0.11$

```
for (i=0; i<n; i++)  
  for (j=0; j<n; j++)  
    A[j][i] = 0;
```

1.65 sec on 10,000 x 10,000 array
 $\approx 15 \times 0.11$

```
p = &A[0][0];  
t = n * n;  
for (i=0; i<t; i++)  
  *p++ = 0;
```

0.11 sec on 10,000 x 10,000 array

A good compiler should know these tradeoffs,
on each target, and generate the best code.
Few real compilers do.

```
bzero((void*) &A[0][0],(size_t) n*n*sizeof(int))
```

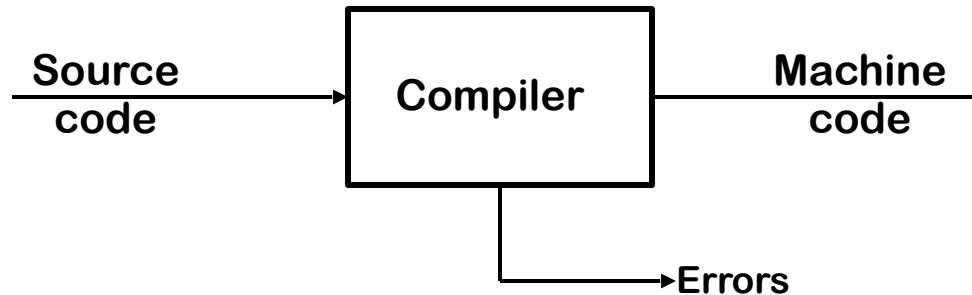
0.52 sec on 10,000 x 10,000 array
 $\approx 5 \times 0.11$

Fundamental Principles of Compilation

- The compiler must preserve the meaning of the program being compiled
- The compiler must improve the input program

The View from 35,000 Feet

High-level View of a Compiler

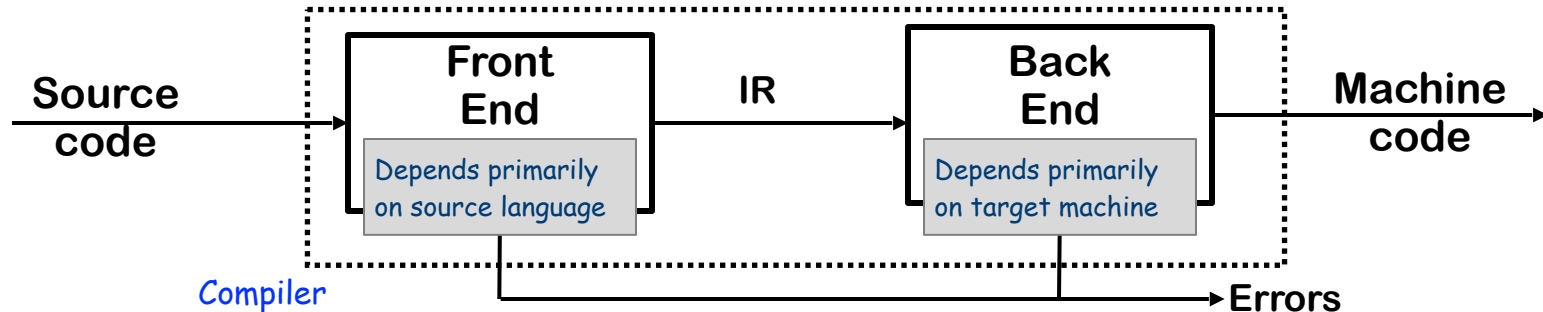


Implications

- Must recognize legal (and illegal) programs
- Must generate correct code
- Must manage storage of all variables (and code)
- Must agree with OS & linker on format for object code

Big step up from assembly language

Traditional Two-pass Compiler



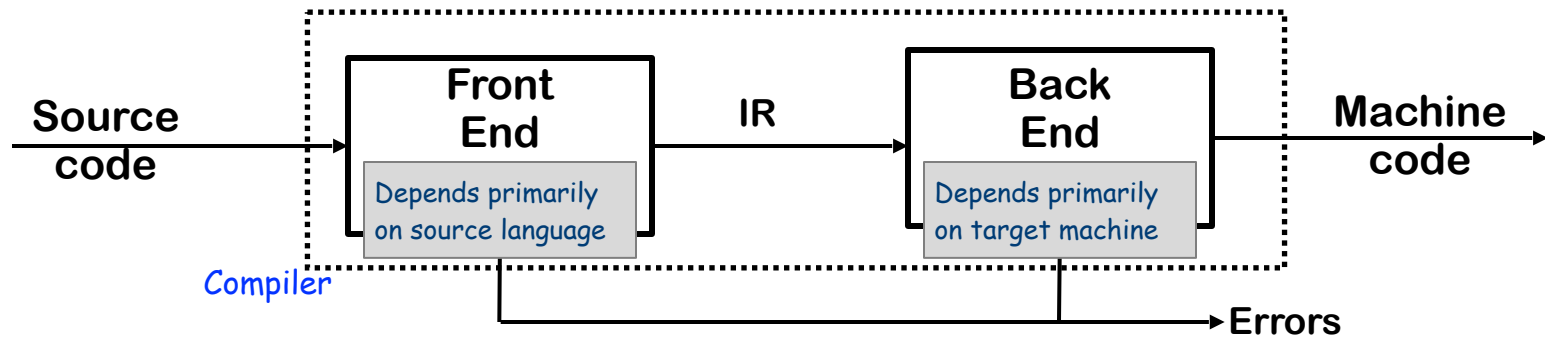
Implications of the division:

- Use an intermediate representation (IR)
- Front end maps legal source code into IR
- Back end maps IR into target machine code
- Admits multiple passes (better code)

Front end is $O(n)$ or $O(n \log n)$

Back end is NP-Complete

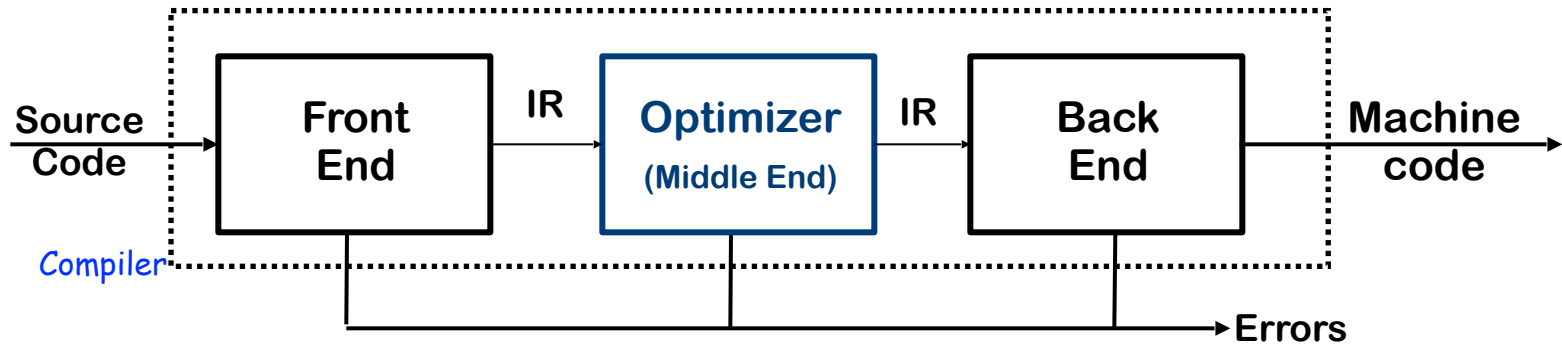
Advantages of two-pass compiler



Classic principle from software engineering:
Separation of concerns

- The some architecture can target a different machine code
- The some architecture can target a source code
- The IR has to encode all the knowledge that the compiler has on the program

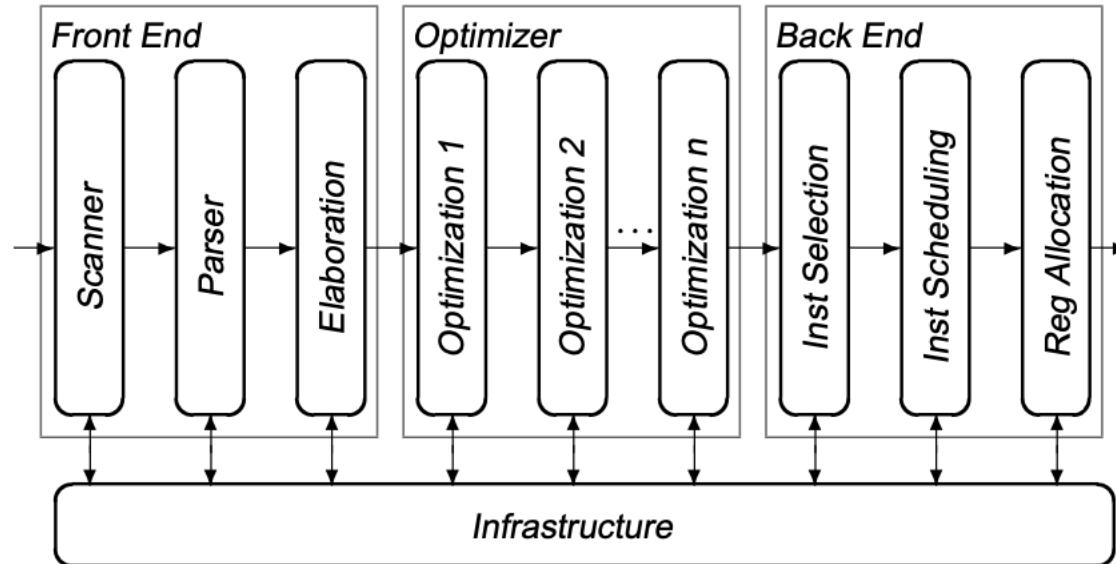
Traditional three-part Compiler



Code Improvement (or Optimization)

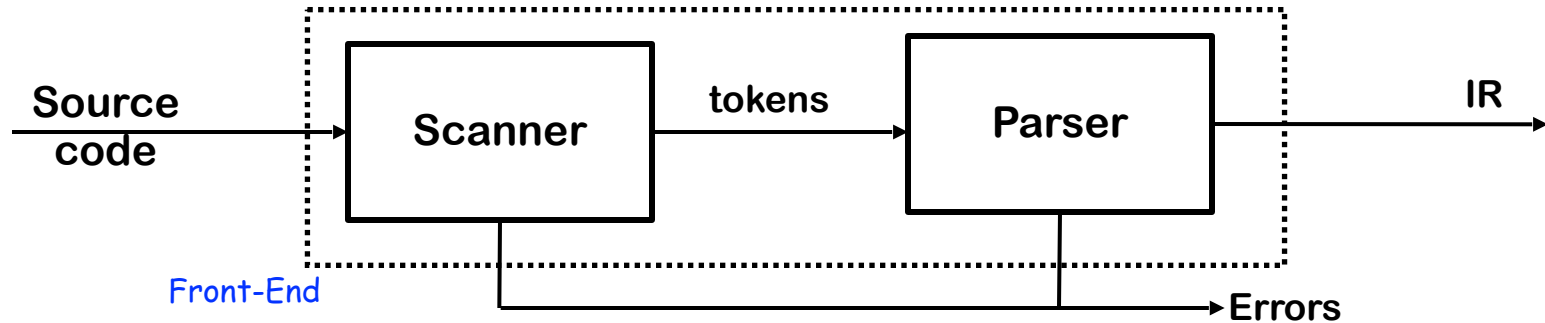
- analyzes IR and transform IR
- primary goal is to reduce running time of the compiled code
 - and/or reduce code space, power consumption, page faults.....
- Must preserve "meaning" of the code

The phases of a Compiler



- In the actual architecture each phase is divided into a series of passes
- The optimiser contains passes that use distinct analyses and transformation to improve the code

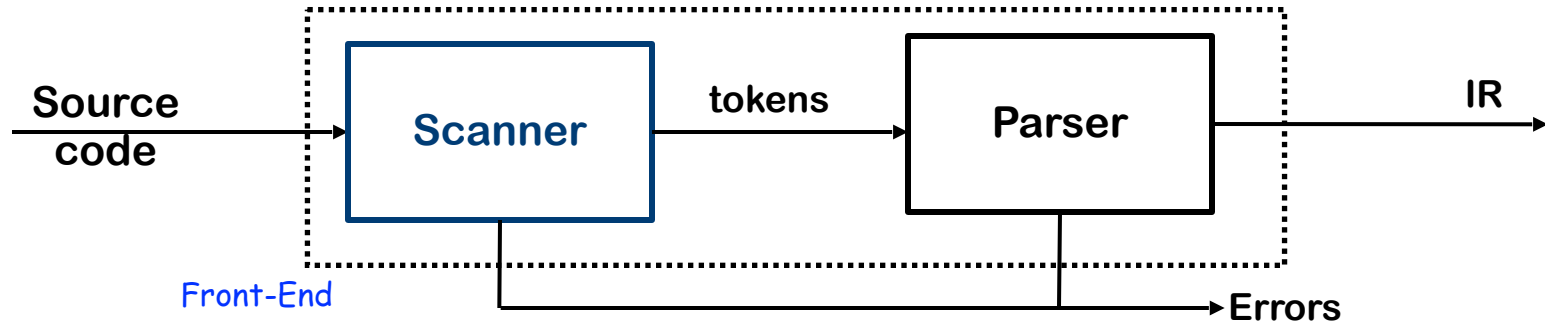
The Front End



Responsibilities

- Recognize legal (& illegal) programs
- Report errors in a useful way
- Produce IR & preliminary storage map
- **Shape** the code for the rest of the compiler
- Much of Front-End construction can be automated

The Front End



Scanner

- Maps stream characters into stream of words (**Lexical analysis**)
- It determines if the stream of characters constitute a legal word
- Produces pairs — a word & its part of speech
 $x = x + y;$ becomes $\langle id, x \rangle \langle op, = \rangle \langle id, x \rangle \langle op, + \rangle \langle id, y \rangle \langle op, ; \rangle$
- Typical words include numbers, identifier, +, -, new, while, if
- Speed is important

Textbooks advocate automatic scanner generation
Commercial practice appears to be hand-coded scanners

Lexical analysis

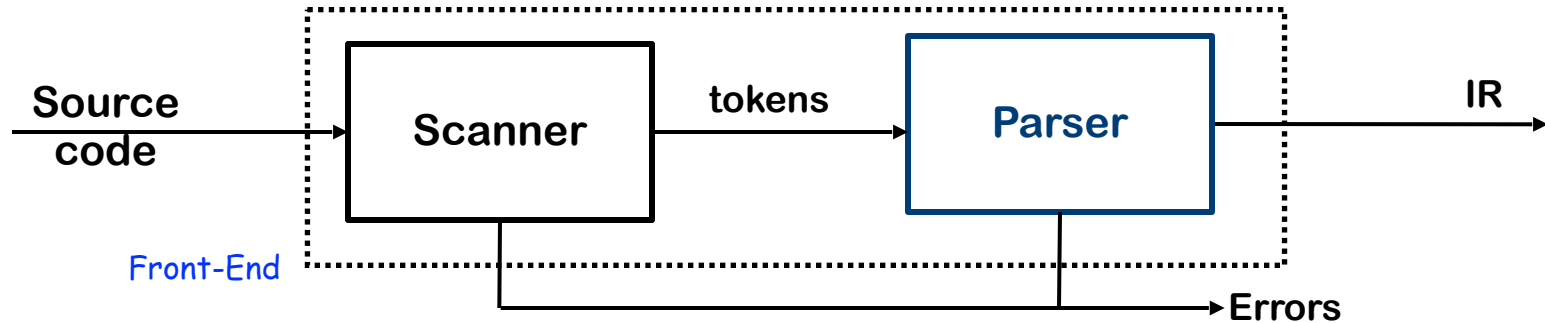
Split program into individual words that makes sense:

1g2h3i is neither a valid identifier nor a valid number

```
while (y < z) {  
    int x = a + b;  
    y += x; }
```

```
T_While  
T_LeftParen  
T_Identifier y  
T_Less  
T_Identifier z  
T_RightParen  
T_OpenBrace  
T_Int  
T_Identifier x  
T_Assign  
T_Identifier a  
T_Plus  
T_Identifier b  
T_Semicolon  
T_Identifier y  
T_PlusAssign  
T_Identifier x  
T_Semicolon  
T_CloseBrace
```

The Front End



Parser

- Check the syntax & reports errors (Syntax Analysis)
- It determines if the stream of words is a sentence in the source language
- Builds IR for source program

Hand-coded parsers are fairly easy to build

Most books advocate using automatic parser generators

Grammars for the Front-End

To recognise words and sentences of the source language, the Front-End uses **grammars** like

$$\begin{array}{l} \text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \\ \quad \quad \quad | \quad \underline{\text{baa}} \end{array}$$

It defines the set of noises that a sheep makes under normal circumstances

It is written in a variant of Backus-Naur Form (BNF)

A grammar $G = (S, N, T, P)$

- S is the start symbol
- N is a set of non-terminal symbols
- T is a set of terminal symbols or words
- P is a set of productions or rewriting rules $(P : N \rightarrow N \cup T)$

Grammars for simple expressions

$S = \textit{Goal}$

$T = \{ \textit{number}, \textit{id}, +, - \}$

$N = \{ \textit{Goal}, \textit{Expr}, \textit{Term}, \textit{Op} \}$

$P = \{ 1, 2, 3, 4, 5, 6, 7 \}$

1. $\textit{Goal} \rightarrow \textit{Expr}$
2. $\textit{Expr} \rightarrow \textit{Expr Op Term}$
3. $\quad \quad \quad / \textit{Term}$
4. $\textit{Term} \rightarrow \textit{number}$
5. $\quad \quad \quad / \textit{id}$
6. $\textit{Op} \rightarrow +$
7. $\quad \quad \quad / -$

- It defines simple expressions with + & - over number and id
- This grammar falls in a class called "context-free grammars", abbreviated CFG

The Front End

Given a CFG, we can *derive* sentences by repeated substitution

1. $Goal \rightarrow Expr$
 2. $Expr \rightarrow Expr Op Term$
 3. | $Term$
 4. $Term \rightarrow \text{number}$
 5. | id
 6. $Op \rightarrow +$
 7. | $-$

<u>Production</u>	<u>Result</u>
	<i>Goal</i>
1	<i>Expr</i>
2	<i>Expr Op Term</i>
5	<i>Expr Op y</i>
7	<i>Expr - y</i>
2	<i>Expr Op term - y</i>
4	<i>Expr Op 2 - y</i>
6	<i>Expr + 2 - y</i>
3	<i>Term + 2 - y</i>
5	<i>x + 2 - y</i>

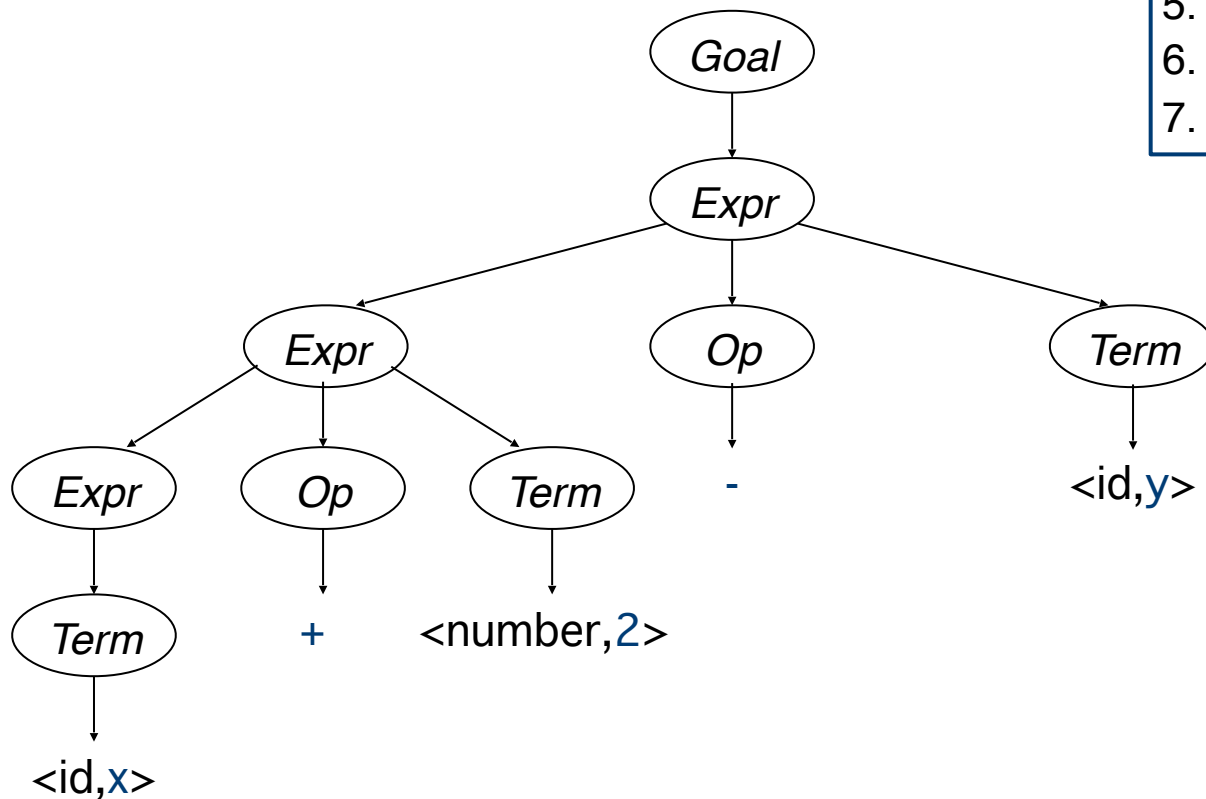
A derivation

To recognize a valid sentence, we reverse this process and start from $x+2-y$

The Front End

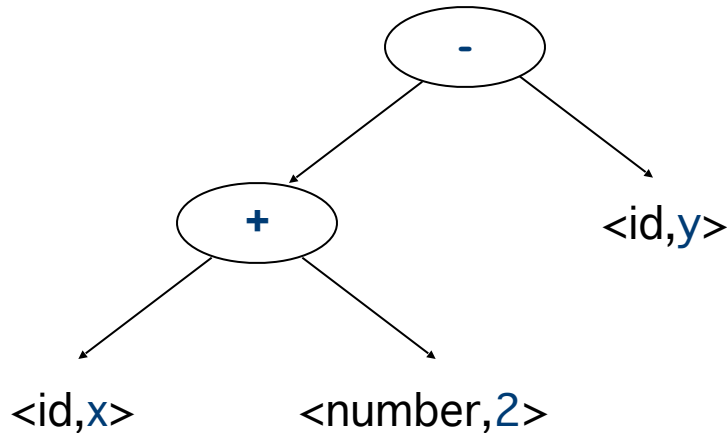
To recognise if $x + 2 - y$
belongs to the language we construct the
parsing tree (or syntax tree)

1. $Goal \rightarrow Expr$
2. $Expr \rightarrow Expr Op Term$
3. | $Term$
4. $Term \rightarrow \text{number}$
5. | id
6. $Op \rightarrow +$
7. | $-$



The Front End

Compilers often use an abstract syntax tree instead of a parse tree



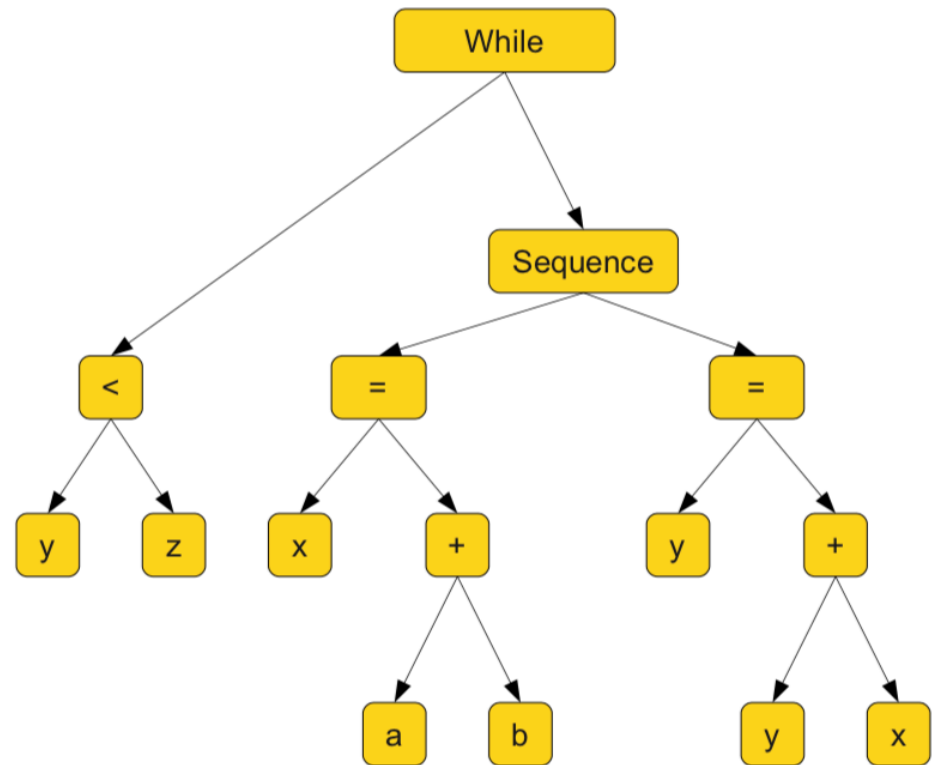
The AST summarizes grammatical structure, without including detail about the derivation

This is much more concise

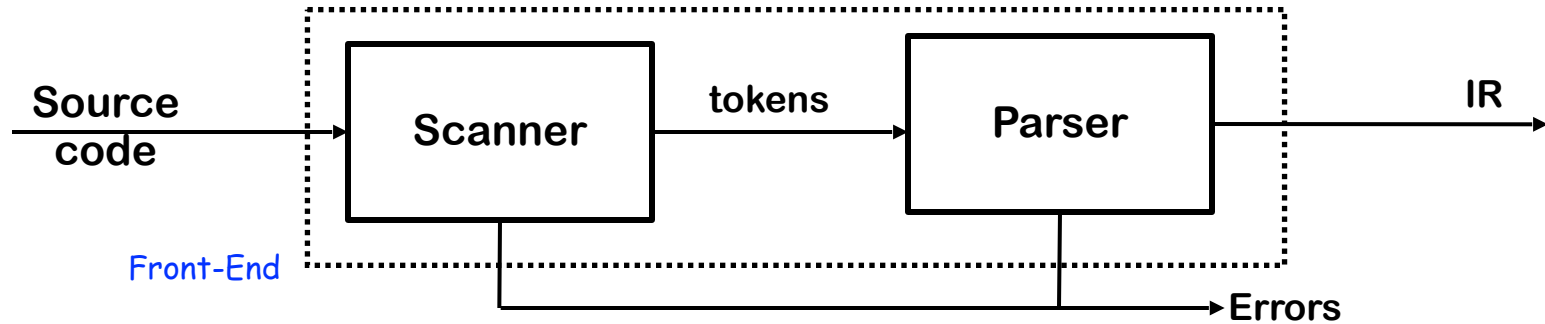
ASTs can be used as intermediate representation

Syntax analysis

```
while (y < z) {  
    int x = a + b;  
    y += x; }
```

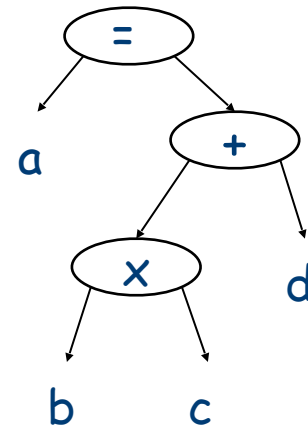


Front-End produces the IR

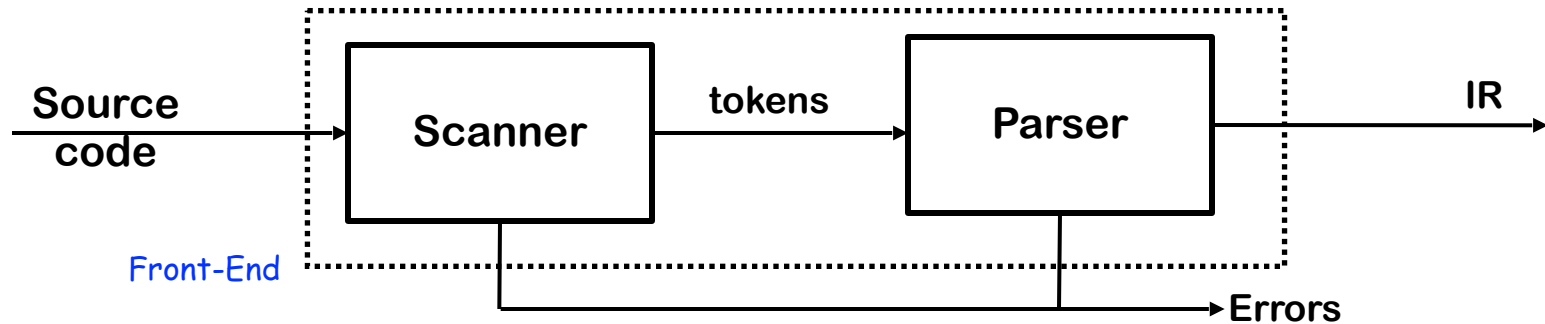


If the IR is the Abstract Syntax Tree

$a = b \times c + d$



Front-End produces the IR



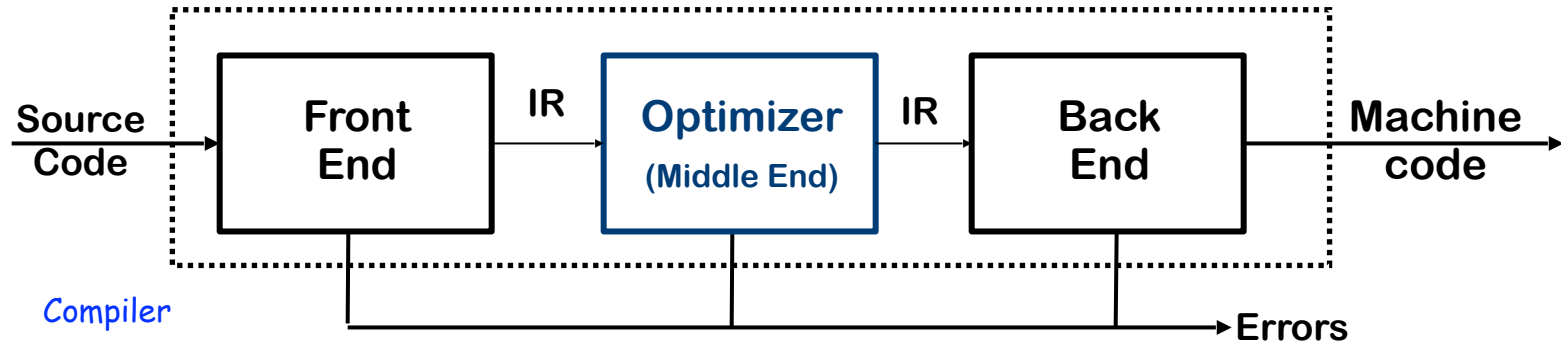
If the IR is the three address code

$a = b \times c + d$



load @b $\Rightarrow$ r_1
load @c $\Rightarrow$ r_2
mult $r_1, r_2 \Rightarrow r_3$
load @d $\Rightarrow$ r_4
add $r_3, r_4 \Rightarrow r_5$
store $r_5 \Rightarrow @a$

The Optimizer



The IR emitted by the Front-End is generated by looking to each statement at the time

The IR program contains code that will work for **any** surrounding context

The optimizer can discover something on the context from the entire IR code and use this knowledge to improve the code

Example of optimizations: loop invariant

b ← ...

c ← ...

a ← 1

for i = 1 to n

 read d

 a ← a × $2 \times b \times c$ × d

end

becomes

b ← ...

c ← ...

a ← 1

t ← 2 × b × c

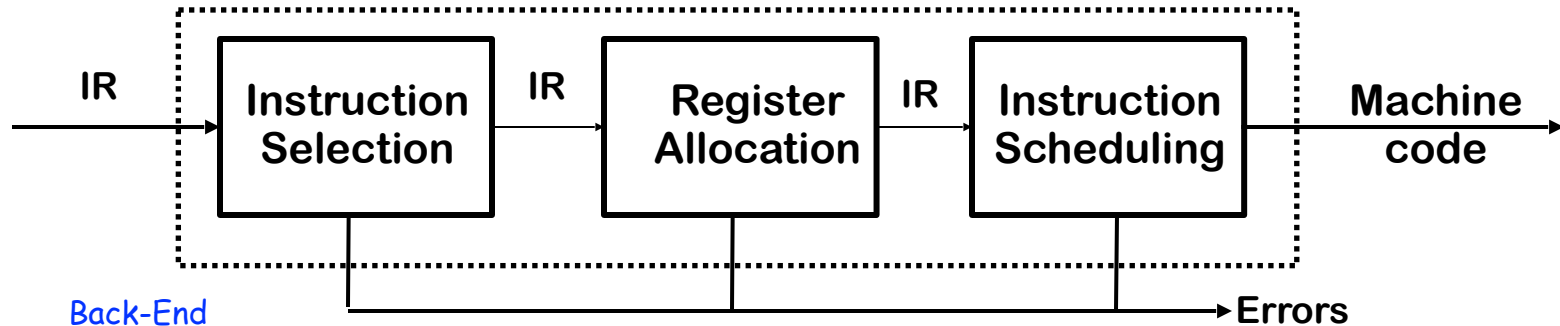
for i = 1 to n

 read d

 a ← a × d × t

end

The Back End



Responsibilities

- Translate IR into target machine code
- Choose instructions to implement each IR operation
- Decide which value to keep in registers
- Reorder the instructions so that efficiency is gained

Automation has been less successful in the back end

About ILOC

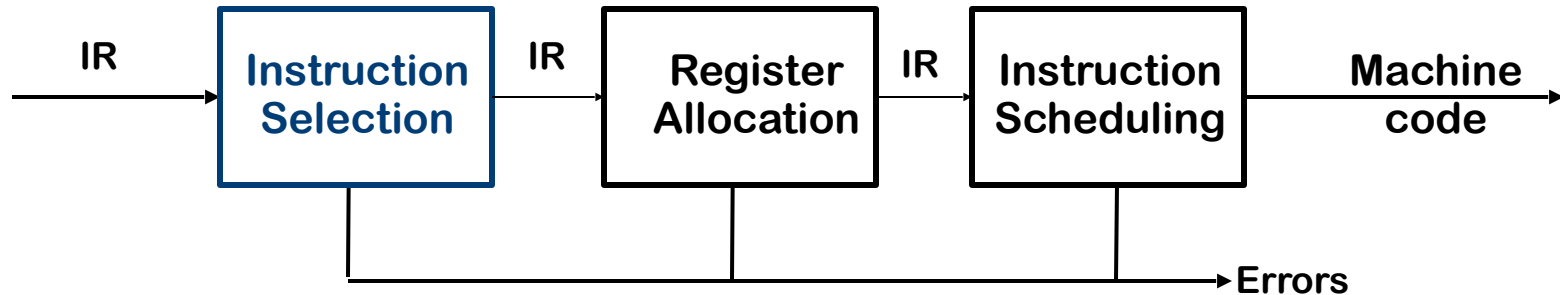
- ILOC (Intermediate Language for Optimizing Compiler) is an assembly language for a simple RISC machine.

ILOC Operation		Meaning
loadAI	$r_1, c_2 \Rightarrow r_3$	$\text{Memory}(r_1 + c_2) \rightarrow r_3$
loadI	$c_1 \Rightarrow r_2$	$c_1 \rightarrow r_2$
mult	$r_1, r_2 \Rightarrow r_3$	$r_1 \times r_2 \rightarrow r_3$
storeAI	$r_1 \Rightarrow r_2, c_3$	$r_1 \rightarrow \text{Memory}(r_2 + c_3)$

Instruction selection for $a = (a \times 2 \times b \times c) \times d$

```
loadAI  rarp, @a ⇒ ra           // load 'a' (Memory(r1+c2) -> r3)
loadI   2        ⇒ r2           // constant 2 into r2 (the constant c1 goes in register r2)
loadAI  rarp, @b ⇒ rb           // load 'b'
loadAI  rarp, @c ⇒ rc           // load 'c'
loadAI  rarp, @d ⇒ rd           // load 'd'
mult    ra, r2  ⇒ ra           // ra ← a × 2
mult    ra, rb  ⇒ ra           // ra ← (a × 2) × b
mult    ra, rc  ⇒ ra           // ra ← (a × 2 × b) × c
mult    ra, rd  ⇒ ra           // ra ← (a × 2 × b × c) × d
storeAI ra      ⇒ rarp, @a       // write ra back to 'a' (r1-> Memory(r2+c3))
```

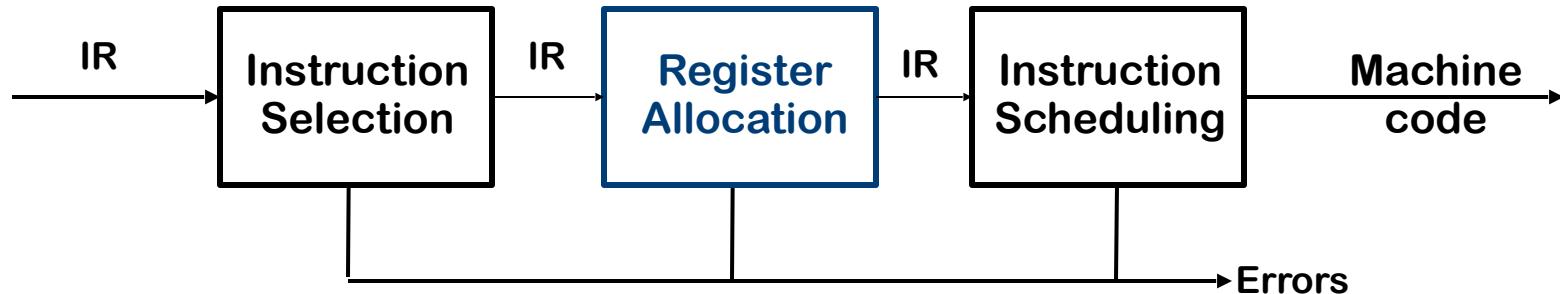
The Back End



Instruction Selection

- It has to translate the IR code into sequence of ISA instructions
- Take advantage of features of the target machine
- Assume an infinite number of (virtual) registers
- Usually viewed as a pattern matching problem
 - ad hoc methods, pattern matching
 - Form of the IR influences choice of technique
 - RISC architecture simplified this problem

The Back End



Register Allocation

- It has to map virtual to physics registers
- Manage a limited set of resources
- Can change instruction choices & insert LOADs & STOREs
- Optimal allocation is NP-Complete in most settings

Compilers approximate solutions to NP-Complete problems

Register allocation for $a = (a \times 2 \times b \times c) \times d$

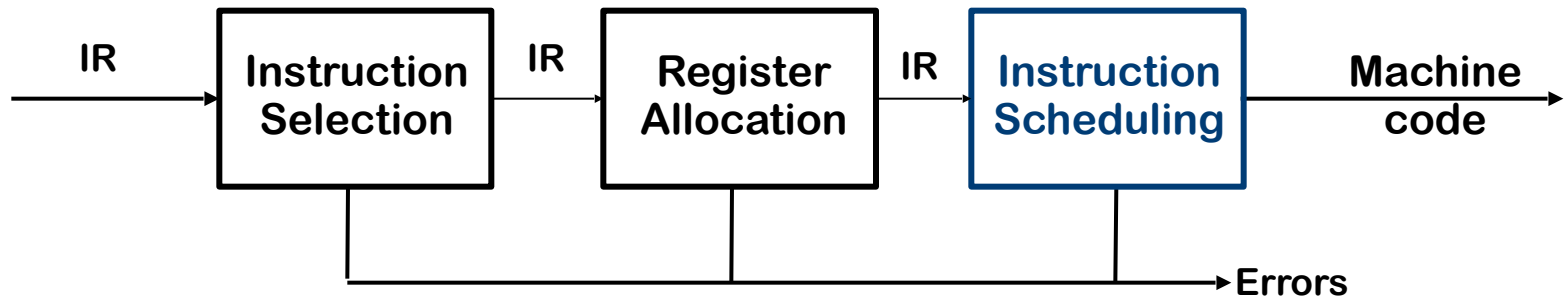
Use 6 registers!

```
loadAI  rarp, @a ⇒ ra      // load 'a' (Memory(r1+c2) → r3)
loadI    2          ⇒ r2    // constant 2 into r2 (the constant c1 goes in register r2)
loadAI  rarp, @b ⇒ rb      // load 'b'
loadAI  rarp, @c ⇒ rc      // load 'c'
loadAI  rarp, @d ⇒ rd      // load 'd'
mult     ra, r2 ⇒ ra      // ra ← a × 2
mult     ra, rb ⇒ ra      // ra ← (a × 2) × b
mult     ra, rc ⇒ ra      // ra ← (a × 2 × b) × c
mult     ra, rd ⇒ ra      // ra ← (a × 2 × b × c) × d
storeAI  ra        ⇒ rarp, @a // write ra back to 'a' (r1 → Memory(r2+c3))
```

Use 3 registers!

```
loadAI  rarp, @a ⇒ r1      // load 'a'
add      r1, r1 ⇒ r1      // r1 ← a × 2
loadAI  rarp, @b ⇒ r2      // load 'b'
mult     r1, r2 ⇒ r1      // r1 ← (a × 2) × b
loadAI  rarp, @c ⇒ r2      // load 'c'
mult     r1, r2 ⇒ r1      // r1 ← (a × 2 × b) × c
loadAI  rarp, @d ⇒ r2      // load 'd'
mult     r1, r2 ⇒ r1      // r1 ← (a × 2 × b × c) × d
storeAI  r1        ⇒ rarp, @a // write ra back to 'a'
```

The Back End



Instruction Scheduling

- It reorder the sequence of instructions to avoid stalls and interlocks
- Use all functional units productively
- Can increase lifetime of variables (changing the allocation)

Optimal scheduling is NP-Complete in nearly all cases

Heuristic techniques are well developed

Before the instruction scheduling

LoadAI, storeAI	3 cycles
mult	2 cycles
others	1 cycle

- The original number of cycles

Start	End			
1	3	loadAI	rarp, @a $\Rightarrow$ r ₁	// load 'a'
4	4	add	r ₁ , r ₁ $\Rightarrow$ r ₁	// r ₁ $\leftarrow$ a $\times$ 2
5	7	loadAI	rarp, @b $\Rightarrow$ r ₂	// load 'b'
8	9	mult	r ₁ , r ₂ $\Rightarrow$ r ₁	// r ₁ $\leftarrow$ (a $\times$ 2) $\times$ b
10	12	loadAI	rarp, @c $\Rightarrow$ r ₂	// load 'c'
13	14	mult	r ₁ , r ₂ $\Rightarrow$ r ₁	// r ₁ $\leftarrow$ (a $\times$ 2 $\times$ b) $\times$ c
15	17	loadAI	rarp, @d $\Rightarrow$ r ₂	// load 'd'
18	19	mult	r ₁ , r ₂ $\Rightarrow$ r ₁	// r ₁ $\leftarrow$ (a $\times$ 2 $\times$ b $\times$ c) $\times$ d
20	22	storeAI	r ₁ $\Rightarrow$ rarp, @a	// write r _a back to 'a'

After the instruction scheduling

Start	End			
1	3	loadAI	$r_{arp}, @a \Rightarrow r_1$	// load 'a'
2	4	loadAI	$r_{arp}, @b \Rightarrow r_2$	// load 'b'
3	5	loadAI	$r_{arp}, @c \Rightarrow r_3$	// load 'c'
4	4	add	$r_1, r_1 \Rightarrow r_1$	// $r_1 \leftarrow a \times 2$
5	6	mult	$r_1, r_2 \Rightarrow r_1$	// $r_1 \leftarrow (a \times 2) \times b$
6	8	loadAI	$r_{arp}, @d \Rightarrow r_2$	// load 'd'
7	8	mult	$r_1, r_3 \Rightarrow r_1$	// $r_1 \leftarrow (a \times 2 \times b) \times c$
9	10	mult	$r_1, r_2 \Rightarrow r_1$	// $r_1 \leftarrow (a \times 2 \times b \times c) \times d$
11	13	storeAI	$r_1 \Rightarrow r_{arp}, @a$	// write r_a back to 'a'